2017-FEB-17

ACME Non-Imaging Data

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The raw non-imaging data will be made available in a spreadsheet format, specifically in a csv format, i.e., with comma-separated values, which can be opened by software such as Microsoft Excel.

The first row of the spreadsheet includes the experiment configuration
… where this is an example from a ground test (presented here in 3 columns rather than 1 long row)

2016-12-19 354 19:17:01:028	MFCA3 Full Scale: 2.000000	IPSU 2: 0
Experiment: 1	EField: 1	IPSU 3: 0
Test Point: Req1.7.9.1_Neg	EFieldPol: 1	IPSU 4: 0
MFCA1 Gas: 0	Igniter: 0	IPSU 5: 1
MFCA1 Full Scale: 20.000000	Burner: 4	IPSU 6: 0
MFCA2 Gas: 0	Thermocouple: 0	PMT 1 Gain: 0.000000
MFCA2 Full Scale: 5.000000	Radiometer: 0	PMT 2 Gain: 0.000000
MFCA3 Gas: 0	IPSU 1: 0	PMT 3 Gain: 0.000000

The second row has these column headings for the data (again shown here in 3 columns rather than a long row)
… where this example shows standard headings (where there are variants, e.g., for BRE measurements)

Time	TCA	RadiometerCenterCJ
MFCA1StaticMixerFlow	TCB	Radiometer-30
MFCA2BypassFlow	TCC	Radiometer-30CJ
MFCA3FuelFlow	TCD	Radiometer-60
IonCurrent	TC200A	Radiometer-60CJ
PTA1	TC200B	RadiometerOffset
PTA2	TCBurner	RadiometerOffsetCJ
PTA3	Radiometer+60	TCE
EfieldVoltage	Radiometer+60CJ	Ch20
PMT1	Radiometer+30	Ch21
PMT2	Radiometer+30CJ	TCCJ
PMT3	RadiometerCenter	EfieldSetpointOut

The data is generally explained through 2 excerpts from the Software IDD, i.e.,
Software Interface Design Description, ACME-DOC-010 Rev C (August 2016)
8. Output Files
4. Internal Interfaces (which includes Tables 15-17*)
**where Table 17 describes channels A0-4 of the DNx-AO-333 analog output board as discussed in 8.1 below.*

8.0 Output FILES

8.1 Data Output File

Data output files are created using the *SystemOpenTestPoint* command which creates a file named and stored at /payload/sci/testpointName_x/testpointName_x_data.csv, where testpointName is the parameter received in the SystemOpenTestPoint command and _x is optional to avoid duplicate files, where x = 1,2,3, etc.

The file contains a two-line header. The first line has the date and time, documentation of the setup received in a *SystemDefineSetup* command, and the PMT gains. The second line has the column headings for the subsequent rows of data. The text of the column headings is defined by the Diagnostics/Header registry entry or the Diagnostics/BreHeader entry depending on whether the *SystemDefineSetup* command experimentType parameter was BRE or not. The column headings should match the data which is the following:

- A timestamp in the format HH:MM:SS:mmm, where mmm is milliseconds.
- 11 channels of analog input from the cube DNx-AI-207 board in the order shown in ***Table 1***.
- 25 channels of analog input from the cube DNx-AI-225 board in the order shown in ***Table 2***. (The names of many of these values change depending on whether it is a BRE test or not, hence the two options for header.)
- Channel AO-4 of the DNx-AO-333 analog output board. This is the E-Field control voltage.

Rows of comma-separated data is added to the file between *DataRecordingStart* and *DataRecordingStop* commands at a rate defined by the Diagnostics/SampleTimeMsec registry entry, see **Error! Reference source not found.**, defaulting to 8 msec for 125 Hz sampling.

8.2 Message Log File

Software messages are logged to files named and stored at /payload/logs/YYYY-MM-DD_hh-mmACME.log, where YYYY is year, MM is month, DD is day of month, hh is hour, and mm is minutes. The date in the name is when the log file is created, soon after software startup.

Each message line in the file begins with a timestamp and a logging category. The timestamp is of the format YYYY-MM-DD DDD hh:mm:ss:mmm, where DDD is day of the year, mmm is milliseconds, and the rest are self-explanatory. See section **Error! Reference source not found.** for information about the logging category and how to filter them out if desired.

8.3 Command Log File

The message log file, section 8.2, includes messages showing commands received and their parameters. Command log files contain only the details of each command received. The files are named and stored at /payload/logs/YYYY-MM-DD_hh-mmACMECommands.log. The format is for each command received a timestamped line of text with the command name followed by timestamped lines of text showing each parameter name and parameter value, including header information.

4.0 INTERNAL INTERFACES

4.1 DNx-AI-207 Board

This analog input layer in the UEI PowerDNA Cube has 16 differential input channels with an input range of +/-10 Volts and 18-bit resolution. Several gains are available but ACME is using a gain of 1.

Table 1 DNx-AI-207 Board Signals

DNx-AI-207 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-0	MFCA1_FLOW	MFCA1 Flow Rate	0 to 5 V	Static mixer/oxygen mass flow controller flow rate. 0 to MFCA1_FULL_SCALE in SLPM. Flow rate (SLPM) = Voltage / 5.0 * MFCA1_FULL_SCALE. Device is Teledyne Hastings HFC-D-302. Full scale depends on installed device.
AIN-1	MFCA2_FLOW	MFCA2 Flow Rate	0 to 5 V	Nitrogen bypass/inert mass flow controller flow rate. 0 to MFCA2_FULL_SCALE in SLPM. Flow rate (SLPM) = Voltage / 5.0 * MFCA2_FULL_SCALE. Device is Teledyne Hastings HFC-D-302. Full scale depends on installed device.
AIN-2	MFCA3_FLOW	MFCA3 Flow Rate	0 to 5 V	Fuel mass flow controller flow rate. 0 to MFCA2_FULL_SCALE in SLPM. Flow rate (SLPM) = Voltage / 5.0 * MFCA2_FULL_SCALE. Device is Teledyne Hastings HFC-D-302. Full scale depends on installed device.
AIN-3	ION_CURRENT	Ion Current	-1.5 to 1.5V	Measurement of ion current between E-Field grid and burner. Ion Current (A) = Voltage/(R_{sense} + R_{DA}), where R_{sense} is resistance of shunt resistor and R_{DA} is the data acquisition resistance. -25 to 25 µA..

DNx-AI-207 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-4	PTA1_OUT	PTA1 Pressure	0.5 to 4.5 V	Outer Coflow pressure. Converted in flight software to 0 to 145 psia using Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4. Device is Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR.
AIN-5	PTA2_OUT	PTA2 Pressure	0.5 to 4.5 V	Inner Coflow/Burner pressure. Converted in flight software to to 0 to 145 psia using Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4. Device is Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR.
AIN-6	PTA3_OUT	PTA3 Pressure	0.5 to 4.5 V	Chamber pressure. Converted in flight software to to 0 to 145 psia using Pressure (psia) = (Voltage – 0.5) * 14.5038 * 10 / 4. Device is Measurement Specialties EPRB-2-VR33-10B-/CM1/ATR.
AIN-7	HV_OUT	E-Field Voltage	-4 to 4 V	Indirect measurement of E-Field high voltage supply. -10000 to +10000 V. E-Field (V) = Voltage * 2486.1795 + 141.9733. (Emperical equation).
AIN-8	PMT1_OUT	Photomultiplier Tube 1 Current	0 to 4 V	0 to 4 µA. Current (µA) = Voltage * 1. Depends on PMT1_GAIN, Table 3. Device is Hamamatsu H10722-10-MOD2. Looks through a 310 nm bandpass interference filter
AIN-9	PMT2_OUT	Photomultiplier Tube 2 Current	0 to 4 V	0 to 4 µA. Current (µA) = Voltage * 1. Depends on PMT2_GAIN, Table 3. Device is Hamamatsu H10722-10-MOD2. ~~Looks through a UV neutral density filter~~ Broadband with a UV-grade fused silica glass window and a neutral density coating. [Stocker clarification]

DNx-AI-207 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-10	PMT3_OUT	Photomultiplier Tube 3 Current	0 to 4 V	0 to 4 µA. Current (µA) = Voltage * 1. Depends on PMT3_GAIN, Table 3. Device is Hamamatsu H10722-10-MOD2. Looks through a 430 nm bandpass interference filter.

4.2 DNx-AI-225 Board

This analog input layer in the UEI PowerDNA Cube has 25 differential input channels with an input range of +/-1.25 Volts and 24-bit resolution. °

Table 2 DNx-AI-225 Board Signals

DNx-AI-225 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-0	TC_A	Rake Thermocouple A	-6.45 to 55 mV	Near or far array Rake A Type K thermocouple, -200 to 1250 C.
	BRE_CTR_180 BRE	BRE Center 180	-6.45 to 55 mV	BRE Center 180 (South) Type K thermocouple, 0 to 500 C.
AIN-1	TC_B	Rake Thermocouple B	0 to 14 mV	Near or far array Rake B Type B thermocouple, 0 to 1700 C.
	BRE_OFF_180 BRE	BRE Offset 180	-6.45 to 55 mV	BRE Offset 180 (Far South) Type K thermocouple, 0 to 500 C.
AIN-2	TC_C	Rake Thermocouple C	0 to 14 mV	Near or far array Rake C Type B thermocouple, 0 to 1700 C.

DNx-AI-225 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
	BRE_CTR [BRE]	BRE Center	-6.45 to 55 mV	BRE Center Type K thermocouple on Medtherm 8-2.688-10SB-0-0-36-21427KSA.
AIN-3	TC_D	Rake Thermocouple D	0 to 14 mV	Near or far array Rake D Type K thermocouple, -200 to 1250 C.
	BRE_OFF [BRE]	BRE Offset	-6.45 to 55 mV	BRE Offset Type K thermocouple on Medtherm 8-2.688-10SB-0-0-36-21427KSA.
AIN-4	TC_200A	200mm Thermocouple A	-6.45 to 55 mV	Igniter side Type K thermocouple, 200 mm from axis. -200 to 1250 C.
AIN-5	TC_200B	200mm Thermocouple B	-6.45 to 55 mV	Relay side Type K thermocouple, 200 mm from axis. -200 to 1250 C.
AIN-6	TC_BNR	Burner Thermocouple	-6.45 to 55 mV	Type K thermocouple on burner. -200 to 1250 C. Not valid when BRE burner is installed.
AIN-7	RA+60	Radiometer +60	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is Dexter Research Center (DRC) XP-0055.
	BRE_OFF+60 [BRE]	+60 BRE Offset	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0249.
AIN-8	RA+60_CJ	Radiometer +60 Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
	BRE_OFF+60CJ [BRE]	+60 BRE Offset Cold Junction		
AIN-9	RA+30	Radiometer +30	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is DRC XP-0055.

DNx-AI-225 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
	BRE_OFF+30 BRE	+30 BRE Offset	0 to 2.5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0249.
AIN-10	RA+30_CJ	Radiometer +30 Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
	BRE_OFF+30CJ BRE	+30 BRE Offset Cold Junction		
AIN-11	RA_CTR	Radiometer Center	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 500. Device is DRC XP-0156.
	BRE_OFF-10 BRE	-10 BRE Offset	0 to 2.5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0249.
AIN-12	RA_CTR_CJ	Radiometer Center Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
	BRE_OFF-10CJ BRE	-10 BRE Offset Cold Junction		
AIN-13	RA-30	Radiometer -30	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is DRC XP-0055.
AIN-14	RA-30_CJ	Radiometer -30 Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.
AIN-15	RA-60	Radiometer -60	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1500. Device is DRC XP-0055.
AIN-16	RA-60_CJ	Radiometer -60 Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is TI LM20.

DNx-AI-225 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AIN-17	RA_OFF	Radiometer Offset	1.25 to 5 V	Thermopile output (V) = (Voltage-1.25) / 1000. Device is DRC XP-0869. This is located on the downstream relay board.
AIN-18	RA_OFF_CJ	Radiometer Offset Cold Junction	0 to 2.5 V	150 to -75 Celcius. 1.57 V at 25C. Device is Texas Instruments LM20.
AIN-19	TC_SPARE	Spare Thermocouple	N/A	To use, an additional thermocouple would need to be hard wired to the Cold Junction Compensation Assembly. This is located on the downstream relay board.
AIN-20	BRE_FLUX_CTR [BRE]	BRE Heat Flux Sensor - Center	0 to 1.25 V	Heat flux sensor output on Medtherm 8-2.688-10SB-0-0-36-21427KSA. Valid only with BRE burner installed. 1mV = 1 Btu/ft^2
AIN-21	BRE_FLUX_OFF [BRE]	BRE Heat Flux Sensor - Offset	0 to 1.25 V	Heat flux sensor output on Medtherm 8-2.688-10SB-0-0-36-21427KSA. Valid only with BRE burner installed. 1mV = 1 Btu/ft^2
AIN-22	N/A	Not Used	N/A	N/A
AIN-23	N/A	Not Used	N/A	N/A
AIN-24	TC_CJ	Thermocouple Cold Junction	0.803 to 1.10 V	Temperature on the Cold Junction Compensation Assembly. 0 to 100 C. Device is Analog Devices AD590. Temperature (Celcius) = (Voltage/0.00295) – 273.15.

[BRE] Valid when the BRE burner is installed.

4.3 DNx-AO-333 Board

This analog output layer in the UEI PowerDNA Cube has 32 channels with an output range of +/-10 Volts and 16-bit resolution.

Table 3 DNx-AO-333 Board

DNx-AO-333 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AO-0	MFCA1_CTRL	MFCA1 Control	0 to 5 V	Static mixer/oxygen mass flow controller setpoint. 0 to MFCA1_FULL_SCALE in SLPM. Full scale depends on which Teledyne Hastings HFC-D-302 is installed. Output voltage = Setpoint(SLPM) * 5 / MFCA1_FULL_SCALE. Or Setpoint (SLPM) = Output voltage * MFCA1_FULL_SCALE / 5. See Table 4.
AO-1	MFCA2_CTRL	MFCA2 Control	0 to 5 V	Nitrogen bypass/inert mass flow controller setpoint. 0 to MFCA2_FULL_SCALE in SLPM. Full scale depends on which Teledyne Hastings HFC-D-302 is installed. Output voltage = Setpoint(SLPM) * 5 / MFCA2_FULL_SCALE. Or Setpoint (SLPM) = Output voltage * MFCA2_FULL_SCALE / 5. See Table 4.
AO-2	MFCA3_CTRL	MFCA3 Control	0 to 5 V	Fuel mass flow controller setpoint. 0 to MFCA3_FULL_SCALE in SLPM. Full scale depends on which Teledyne Hastings HFC-D-302 is installed. Output voltage = Setpoint(SLPM) * 5 / MFCA3_FULL_SCALE. Or Setpoint (SLPM) = Output voltage * MFCA3_FULL_SCALE / 5. See Table 4.
AO-3	PMT_PWR	PMT Power	0 or 5 V	Power to three photomultipler tubes. 0 V = off. 5 V = on.
AO-4	HV_CTRL	E-Field HV Control	0 to 10 V	Sets E-field HV supply output between about -10000 and 10000 V. EField See EFieldSetVoltage command, section **Error! Reference source not found.**, for equation..
AO-5	HV_EN	E-Field HV Enable	0 or 5 V	E-Field high voltage supply enable. 0 V = disabled, 5 V = enabled
AO-6	N/A	Not Used	0 or 5 V	N/A

DNx-AO-333 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AO-7	IGN_EN	Igniter Enable	0 or 5 V	0 V = disabled. 5 V enabled. Software sets IGN_CTRL then pulses IGN_EN.
AO-8	IGN_CTRL	Igniter Control	-0.6 to +0.8V	Sets voltage at the igniter between 0.46 to 12 V.
AO-9	MFCA1_PWR	MFCA1 Power	0 or 5 V	Power to static mixer/oxygen MFC. 0 V = unpowered. 5 V = powered.
AO-10	MFCA2_PWR	MFCA2 Power	0 or 5 V	Power to nitrogen bypass/inert MFC. 0 V = unpowered. 5 V = powered.
AO-11	MFCA3_PWR	MFCA3 Power	0 or 5 V	Power to fuel MFC. 0 V = unpowered. 5 V = powered.
AO-12	RB_UP_EN	Relay Board Upstream Enable	0 or 5 V	0 V = disables signals. 5 V = state of signal output depends on signal value. Applicable signals: SV1, SV2, SV3, SV4, SV5, SV6, SV7, SV8, SV9, PMT_PWR, MFCA1_PWR, MFCA2_PWR, MFCA3_PWR
AO-13	TFP_MOTOR_RST	TFP Motor Reset	0 or 5 V	0 V to rest TFP stepper motor. 5 V to enable motor.
AO-14	IGN_MOTOR_RST	Igniter Motor Reset	0 or 5 V	0 V to reset igniter stepper motor. 5 V to enable motor.
AO-15	SVA4	Solenoid Valve A4	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-16	SVA5	Solenoid Valve A5	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-17	SVA6	Solenoid Valve A6	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.

DNx-AO-333 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AO-18	SVA7	Solenoid Valve A7	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-19	SVA8	Solenoid Valve A8	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-20	SVA9	Solenoid Valve A9	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-21	SVA1	Solenoid Valve A1	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-22	SVA2	Solenoid Valve A2	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-23	SVA3	Solenoid Valve A3	0 or 5 V	0 V = no power to solenoid valve. 5 V = power solenoid valve, opening this normally closed valve.
AO-24	N/A	Not Used	0 or 5 V	N/A
AO-25	PMT1_GAIN	PMT1 Gain	4.63 to 10 V	For recommended device control voltage of 0.5 to 1.1 V. Gain range is $5x10^3$ to $3.5x10^6$. Gain=$2x10^6$ * (Output voltage/9.2645)^8.643856. Or Output voltage = 10^(0.115689 * log(Gain) + 0.23786213). Device is Hamamatsu H10722-10-MOD2.
AO-26	PMT2_GAIN	PMT2 Gain	4.63 to 10 V	For recommended device control voltage of 0.5 to 1.1 V. Gain range is $5x10^3$ to $3.5x10^6$. Gain=$2x10^6$ * (Output voltage/9.2645)^8.643856. Or Output voltage = 10^(0.115689 * log(Gain) + 0.23786213). Device is Hamamatsu H10722-10-MOD2.

DNx-AO-333 Board Signals				
Channel	Mnemonic	Name	Signal Range	Description, Conversion
AO-27	PMT3_GAIN	PMT3 Gain	4.63 to 10	For recommended device control voltage of 0.5 to 1.1 V. Gain range is $5x10^3$ to $3.5x10^6$. Gain=$2x10^6$ * (Output voltage/9.2645)^8.643856. Or Output voltage = 10^(0.115689 * log(Gain) + 0.23786213). Device is Hamamatsu H10722-10-MOD2.
AO-28	LED_SIDE	LED Side Bank	0 or 5 V	Bank of 3 white LEDs to side of analog camera turning mirror. 0 V = off. 5 V = on.
AO-29	LED_END	LED End Bank	0 or 5 V	Bank of 3 white LEDs past the end of the analog camera turning mirror. 0 V = off. 5 V = on.
AO-30	RB_DOWN_EN	Relay Board Downstream Enable	0 or 5 V	0 V = disables signals. 5 V = state of signal output depends on signal value. Applicable signals: HV1, HV2, LED1, LED2, TFP_RST, IGN_RST, IGN_EN
AO-31	N/A	Not Used	0 or 5 V	N/A

4.4 Serial Communication

Table 4 Serial Communication

Serial Communication				
Source	Device Name	Mnemonic	Signal Range	Description
Lippert SBC	/dev/ttyS0	MFCA1_FULL_SCALE	Depends on what MFC is installed. 0.5, 20 SLPM.	Full scale flow of static mixer/oxygen mass flow controller in SLPM. Pulled from gas list item 18 in the result of the SFL (System Full List) command to device.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA2_FULL_SCALE	Depends on what MFC is installed. 0.01, 0.02, 0.05, 0.1, 0.5, 2, 5 SLPM.	Full scale flow of nitrogen bypass/Inert mass flow controller in SLPM. Pulled from gas list item 18 in the result of the SFL (System Full List) command to device.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA3_FULL_SCALE	Depends on what MFC is installed. 0.05, 0.1, 0.2, 0.5, 1 SLPM.	Full scale flow of fuel mass flow controller in SLPM. Pulled from gas list item 18 in the result of the SFL (System Full List) command to device.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA1_STATE	0-10. See **Error! Reference source not found.**.	State of the oxygen/static mixer mass flow controller. Result of a MS (get state) command.
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA2_STATE	0-10. See **Error! Reference source not found.**.	State of the nitrogen bypass mass flow controller. Result of a MS (get state) command.

Serial Communication				
Source	Device Name	Mnemonic	Signal Range	Description
Lippert SBC RXD1/TXD1	/dev/ttyS0	MFCA3_STATE	0-10. See **Error! Reference source not found.**.	State of the fuel mass flow controller. Result of an MS (get state) command.
ConnectTech serial card P5	/dev/CTI4	CUBE_SERIAL	See DNA-PPCx-1G PowerDNA Gigabit Ethernet I/O Cube User Manual.	UEI cube serial connection. Can be used to retrieve version numbers, set IP addresses, and update firmware.
ConnectTech serial card P6	/dev/CTI5	ANALOG_CAM_AGC_MODE	On or Off.	Turn automatic gain control on or off
ConnectTech serial card P6	/dev/CTI5	ANALOG_CAM_GAIN	6, 12, 21, and 31 dB.	Set the automatic gain control gain.
ConnectTech serial card P6	/dev/CTI5	ANALOG_CAM_SHUTTER	1/60, 1/100, 1/250, 1/500, 1/1000, 1/2000, 1/4000, 1/10000, 1/20000, 1/30000 seconds.	Set the shutter speed.
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	IGN_MOTOR_SET_POS	In counts. 0 is the retracted position. Maximum extended position depends on registry values Igniter/IgniterLimits and Igniter/IgniterMaxCount and the burner type last specified by a System Define Setup command. Default maximum is 2500 counts.	Axis 1 control of motion control board in AVP controlling igniter motor in CIA.

Serial Communication				
Source	Device Name	Mnemonic	Signal Range	Description
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	IGN_MOTOR_GET_POS	See Signal Range for IGN_MOTION_SET_POS	Axis 1 position reporting of motion control board in AVP controlling igniter motor in CIA.
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	TFP_ MOTOR _SET_POS	In counts. 0=retracted. Maximum=closer to burner. Maximum depends on burner type and registry values TfpMaxCount and TfpToBurnerOffsets. See Set TFP Position command for calculation of maximum.	Axis 2 control of motion control board in AVP controlling TFP (thin filament pyrometry array) motor in CIA.
ConnectTech serial card P7	/dev/CTI6 and /dev/ttyS2	TFP_ MOTOR _GET_POS	See Signal Range for TFP_MOTION_SET_POS	Axis 2 position reporting of motion control board in AVP controlling TFP motor in CIA.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	ZOOM_ MOTOR _SET_POS	0-85000 counts. 0 is zoomed out. 85000 is zoomed in.	Not valid when digital camera is connected to the IPSU-G. Axis 1 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	ZOOM_ MOTOR _GET_POS	0-85000 counts. 0 is zoomed out. 85000 is zoomed in.	Not valid when digital camera is connected to the IPSU-G. Axis 1 position reporting of motion control board in digital camera package.

Serial Communication				
Source	Device Name	Mnemonic	Signal Range	Description
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	IRIS_ MOTOR _SET_POS	0-76000 counts. 0 is open. 76000 is closed.	Not valid when digital camera is connected to the IPSU-G. Axis 2 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	IRIS_ MOTOR _GET_POS	0-76000 counts. 0 is open. 76000 is closed.	Not valid when digital camera is connected to the IPSU-G. Axis 2 position reporting of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FOCUS_ MOTOR _SET_POS	0-101000 counts. 0 is near focus. 101000 is far focus.	Not valid when digital camera is connected to the IPSU-G. Axis 3 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FOCUS_ MOTOR _GET_POS	0-101000 counts. 0 is near focus. 101000 is far focus.	Not valid when digital camera is connected to the IPSU-G. Axis 3 position reporting of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FILTER_ MOTOR _SET_POS	0-451000 counts. Filters at 0, 120000, 240000, and 360000 counts.	Not valid when digital camera is connected to the IPSU-G. Axis 4 control of motion control board in digital camera package.
ConnectTech serial card P8	/det/CTI7 and /dev/ttyS3	FILTER_ MOTOR _GET_POS	0-451000 counts. Filters at 0, 120000, 240000, and 360000 counts.	Not valid when digital camera is connected to the IPSU-G. Axis 4 position reporting of motion control board in digital camera package.

4.5 Internal Ethernet Interfaces

See also **Error! Reference source not found.**, External Ethernet Interfaces where the interface to the IOP is listed. The 'Kernels' string points to a hardware ethernet interface on the processor. It is used in /etc/udev/rules.d/90-net.rules to ensure that the device name is always assigned to the same connector.

Table 5 Internal Ethernet Interfaces

Source	Device Name	IP Address	Net Mask	Kernels	Description
Lippert SBC	eth1	192.168.10.1	255.255.0.0	0000:06:00.0	Interface to data camera when connected directly to ACME and not IPSU-G.
Lippert SBC	eth2	192.168.100.1	255.255.255.0	0000:08:00.0	Interface to UEI Cube.
ACME Camera Package	DeviceIPAddress	192.168.10.100	255.255.0.0		Camera IP address. Changeable using the CLIpConfig command line utility included in ACME software install.
UEI Cube	ip2	192.168.100.102	255.255.255.0		Cube main ethernet port. Labeled NIC1. (yes, for ip2)
UEI Cube	ip	192.168.100.2	255.255.255.0		Cube secondary/diagnostic ethernet port. Labeled NIC2. NOT CONNECTED.